

一、仿射空间中的张量分析

①

1.1 n 维空间点坐标为 $x^\mu = (x^1, x^2, \dots, x^n)$

x^μ 与 $\tilde{x}^\mu$ 之间坐标变换 $\tilde{x}^\mu = \tilde{x}^\mu(x)$

坐标微分变换 $d\tilde{x}^\mu = \frac{\partial \tilde{x}^\mu}{\partial x^\alpha} dx^\alpha$ 对 α 求和 $\rightarrow$ 爱因斯坦约定。
变换矩阵 $\det \left| \frac{\partial \tilde{x}^\mu}{\partial x^\alpha} \right| \neq 0$ 或 ∞ 则存在逆变换

$$dx^\alpha = \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} d\tilde{x}^\mu$$

两变换矩阵满足

$$\begin{cases} \frac{\partial \tilde{x}^\mu}{\partial x^\alpha} \frac{\partial x^\alpha}{\partial \tilde{x}^\nu} = \delta_\nu^\mu = \begin{cases} 1, & \mu = \nu \\ 0, & \mu \neq \nu \end{cases} \\ \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \frac{\partial \tilde{x}^\mu}{\partial x^\beta} = \delta_\beta^\alpha \end{cases}$$

逆变张量 (上标)

零阶 $\rightarrow$ 标量 (a) 有 n^0 个分量, 坐标 x^μ 下值为 T

(b) $x^\mu \rightarrow \tilde{x}^\mu, T \rightarrow \tilde{T}, \tilde{T}(\tilde{x}) = T(x)$ 标量的值在坐标变换下保持不变。

-阶 $\rightarrow$ 矢量 (a) 有 n^1 个分量, x^μ 下记为 T^μ

(b) $x^\mu \rightarrow \tilde{x}^\mu, T^\mu \rightarrow \tilde{T}^\mu, \tilde{T}^\mu = \frac{\partial \tilde{x}^\mu}{\partial x^\alpha} T^\alpha$ (变换与坐标微分一致)

=阶 $\rightarrow$ 张量 (a) 有 n^2 个分量, x^μ 下记为 $T^{\mu\nu}$

(b) $x^\mu \rightarrow \tilde{x}^\mu, T^{\mu\nu} \rightarrow \tilde{T}^{\mu\nu}, \tilde{T}^{\mu\nu} = \frac{\partial \tilde{x}^\mu}{\partial x^\alpha} \frac{\partial \tilde{x}^\nu}{\partial x^\beta} T^{\alpha\beta}$

协变张量 (下标)

零阶无区别 -阶 $x^\mu \rightarrow \tilde{x}^\mu, T_\mu \rightarrow \tilde{T}_\mu, \tilde{T}_\mu = \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} T_\alpha$

=阶 $x^\mu \rightarrow \tilde{x}^\mu, T_{\mu\nu} \rightarrow \tilde{T}_{\mu\nu}, \tilde{T}_{\mu\nu} = \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \frac{\partial x^\beta}{\partial \tilde{x}^\nu} T_{\alpha\beta}$

混合张量 (p, q) 阶为 $T^{\alpha_1 \alpha_2 \dots \alpha_p}_{\beta_1 \beta_2 \dots \beta_q}$, =阶混合为例 $\tilde{T}^\mu_\nu = \frac{\partial \tilde{x}^\mu}{\partial x^\alpha} \frac{\partial x^\beta}{\partial \tilde{x}^\nu} T^\alpha_\beta$



张量间运算

加减法 $C^\mu_\nu = A^\mu_\nu \pm B^\mu_\nu$

乘法(外乘) $C^{\mu\nu}_\lambda = A^\mu_\lambda B^\nu$

缩并 $C^\nu = A^{\lambda\nu}_\lambda$ 或 $D^\mu = A^{\mu\lambda}_\lambda$

矢量内积 $C = A^\mu B_\mu$ (外乘后缩并)

1.2 张量的对称性

二阶逆变~用矩阵导出

$$T^{\mu\nu} = \begin{pmatrix} T^{11} & T^{12} & \dots & T^{1n} \\ T^{21} & & & \\ \vdots & & & \\ T^{nn} & & & \end{pmatrix}$$

若对称, $T^{\mu\nu} = T^{\nu\mu}$, 指标 μ, ν 对称

反对称, $T^{\mu\nu} = -T^{\nu\mu}$, 指标 μ, ν 反对称

与坐标无关

协变逆用,
变换坐标后不保持 $\leftarrow$ 混合不适用

不对称的二阶逆变/协变可分解为对称与反对称

$$T^{\mu\nu} = S^{\mu\nu} + A^{\mu\nu} \quad \text{且} \quad \begin{cases} S^{\mu\nu} = S^{\nu\mu} \\ A^{\mu\nu} = -A^{\nu\mu} \end{cases}$$

唯一确定分解法为

$$\begin{cases} S^{\mu\nu} = \frac{1}{2}(T^{\mu\nu} + T^{\nu\mu}) \equiv T^{(\mu\nu)} \\ A^{\mu\nu} = \frac{1}{2}(T^{\mu\nu} - T^{\nu\mu}) \equiv T^{[\mu\nu]} \end{cases}$$

反对称张量有一些好的性质

(a) $\mu=\nu$ 时, $T^{\mu\mu} = 0$

(b) n -dim 中最高阶反对称为 n -dim, 并仅有一个独立分量

(c) n -dim 中 $(n-1)$ -dim 反对称只有 n 个独立分量

1.3 矢量平移和仿射联络

平移($P \rightarrow Q$): $x^\mu \rightarrow x^\mu + dx^\mu$

$P: A_\mu(P) \rightarrow Q: A_\mu(P \rightarrow Q)$ 引起改变 $\delta A_\mu(P) \propto A_\mu(P)$

显然有 $\delta A_\mu(P) = A_\mu(P \rightarrow Q) - A_\mu(P)$

$$= \Gamma^\lambda_{\mu\nu}(P) A_\lambda(P) dx^\nu$$

系数 $\Gamma^\lambda_{\mu\nu}(P)$ 为 P 点的仿射联络

要求协变性性质,

$$\begin{aligned} \tilde{A}_\mu(P \rightarrow Q) &= \left(\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \right)_Q A_\alpha(P \rightarrow Q) \\ \text{变换矩阵有 } \left(\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \right)_Q &= \left(\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \right)_P + \left(\frac{\partial^2 x^\alpha}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \right)_P d\tilde{x}^\nu \\ &= \left(\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \right)_P + \left(\frac{\partial^2 x^\alpha}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} \right) dx^\sigma \end{aligned}$$

变为

$$\tilde{A}_\mu + \Gamma^\lambda_{\mu\nu} \tilde{A}_\lambda d\tilde{x}^\nu = \left(\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} + \frac{\partial^2 x^\alpha}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} dx^\sigma \right) (A_\alpha + \Gamma^\beta_{\alpha\sigma} A_\beta dx^\sigma)$$

式中所有量的自变量均为 P 的坐标

有 $\tilde{A}_\mu = \frac{\partial x^\beta}{\partial \tilde{x}^\mu} A_\beta$, $d\tilde{x}^\nu = \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} dx^\sigma$, 代入(反面推导)

$$\begin{aligned} &= \Gamma^\lambda_{\mu\nu} \frac{\partial x^\beta}{\partial \tilde{x}^\lambda} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} = \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\lambda} \frac{\partial \tilde{x}^\lambda}{\partial x^\sigma} + \Gamma^\beta_{\alpha\sigma} \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \\ &+ \Gamma^\lambda_{\mu\nu} = \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\lambda}{\partial x^\beta} + \Gamma^\beta_{\alpha\sigma} \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \frac{\partial \tilde{x}^\sigma}{\partial x^\beta} \frac{\partial \tilde{x}^\lambda}{\partial x^\beta} \end{aligned}$$



$$\tilde{A}_\mu + \tilde{\Gamma}_{\mu\nu}^\lambda \tilde{A}_\lambda d\tilde{x}^\nu = \left(\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} + \frac{\partial^2 x^\alpha}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} dx^\sigma \right) * (A_\alpha + \Gamma_{\alpha\sigma}^\beta A_\beta dx^\sigma)$$

$$L = \cancel{\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} A_\alpha} + \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \Gamma_{\alpha\sigma}^\beta A_\beta dx^\sigma + \frac{\partial^2 x^\alpha}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} dx^\sigma A_\alpha + \frac{\partial^2 x^\alpha}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} \underbrace{dx^\sigma}_{\text{微分}} \underbrace{\Gamma_{\alpha\sigma}^\beta A_\beta}_{\text{微分}}$$

$$L = \cancel{\frac{\partial x^\alpha}{\partial \tilde{x}^\mu} A_\alpha} + \tilde{\Gamma}_{\mu\nu}^\lambda \frac{\partial x^\beta}{\partial \tilde{x}^\lambda} A_\beta \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} dx^\sigma = \text{微分}$$

$$\ddagger \quad \tilde{\Gamma}_{\mu\nu}^\lambda \frac{\partial x^\beta}{\partial \tilde{x}^\lambda} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} \cancel{A_\beta dx^\sigma} = \Gamma_{\alpha\sigma}^\beta \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \cancel{A_\beta dx^\sigma} + \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} \cancel{A_\beta dx^\sigma} \quad \text{合法换标}$$

$$\ddagger \quad \tilde{\Gamma}_{\mu\nu}^\lambda \frac{\partial x^\beta}{\partial \tilde{x}^\lambda} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma} = \Gamma_{\alpha\sigma}^\beta \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} + \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\sigma}$$

$$\ddagger \quad \underset{\blacktriangle}{\tilde{\Gamma}_{\mu\nu}^\lambda} = \frac{\partial^2 x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\nu}{\partial x^\beta} + \underset{\blacktriangle}{\Gamma_{\alpha\sigma}^\beta} \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} \frac{\partial \tilde{x}^\lambda}{\partial x^\beta}$$



以上以协变为例, 对于逆变同理

⑤

$$\text{有 } A^\mu(P \rightarrow Q) = A^\mu + \delta A^\mu = A^\mu - \Gamma_{\lambda\nu}^\mu A^\lambda dx^\nu$$

仿射联络的性质:

(a) $T_{\mu\nu}^\lambda \equiv \Gamma_{\mu\nu}^\lambda - \Gamma_{\nu\mu}^\lambda = (1,2)$ 阶张量

(b) 联络 $\Gamma_{\mu\nu}^\lambda$ 对 μ, ν 不对称, 那么 $\tilde{\Gamma}_{\mu\nu}^\lambda \equiv \Gamma_{\nu\mu}^\lambda$ 也是一种联络

(c) 联络的对称组合 $\bar{\Gamma}_{\mu\nu}^\lambda \equiv \frac{1}{2}(\Gamma_{\mu\nu}^\lambda + \Gamma_{\nu\mu}^\lambda)$ 也是一种联络, 为对称联络 (对 ν, μ 对称)

(d) 反对称组合 $\bar{\Gamma}_{[\mu\nu]}^\lambda \equiv \frac{1}{2}(\Gamma_{\mu\nu}^\lambda - \Gamma_{\nu\mu}^\lambda) \rightarrow$ 仿射空间的挠率张量

(e) 非对称联络总可表示为 $\Gamma_{\mu\nu}^\lambda = \bar{\Gamma}_{(\mu\nu)}^\lambda + \bar{\Gamma}_{[\mu\nu]}^\lambda$

1.4 张量的协变微商

(1) 标量场的微商

$$T_{, \mu} \equiv \partial T / \partial x^\mu$$

变换, $T_{, \mu} \rightarrow \tilde{T}_{, \mu} \equiv \frac{\partial \tilde{T}}{\partial \tilde{x}^\mu} = \frac{\partial T}{\partial x^\alpha} \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} = T_{, \alpha} \frac{\partial x^\alpha}{\partial \tilde{x}^\mu}$ 表明 $T_{, \mu}$ 是协变矢量

协变微商 $T_{; \mu} = T_{, \mu}$

(2) 协变矢量场的微商, 场 $T_\mu(x)$

$$T_{\mu; \nu}(x) \equiv \partial T_\mu(x) / \partial x^\nu = \lim_{Q \rightarrow P} \frac{T_\mu(Q) - T_\mu(P)}{\Delta x^\nu}$$

利用平移操作 $\lim_{Q \rightarrow P} \frac{T_\mu(Q) - T_\mu(P \rightarrow Q)}{\Delta x^\nu} = T_{\mu; \nu} \quad (0,2)$ 阶

平移公式给出 $T_{\mu; \nu} = T_{\mu, \nu} - \Gamma_{\mu\nu}^\lambda T_\lambda \rightarrow$ 协变微商公式 右边两项均非张量, 但差是张量

乘法法则 $(A^{\dots} B^{\dots})_{; \lambda} = (A^{\dots})_{; \lambda} (B^{\dots}) + (A^{\dots})(B^{\dots})_{; \lambda}$

(3) 逆变矢量场 $A^\mu(x)$

考虑 A^μ 与 B_μ 可构成标量 $A^\mu B_\mu$

那么 $(A^\mu B_\mu)_{; \lambda} = (A^\mu B_\mu)_{, \lambda} \quad \Rightarrow \quad A^\mu_{; \lambda} B_\mu + A^\mu B_{\mu; \lambda} = A^\mu_{, \lambda} B_\mu + A^\mu B_{\mu, \lambda}$

~~而 $B_{\mu; \lambda} = B_{\mu, \lambda} - \Gamma_{\alpha\lambda}^\mu B_\mu$~~ 而 $B_{\mu; \lambda} = B_{\mu, \lambda} - \Gamma_{\mu\lambda}^\nu B_\nu \quad \Rightarrow \quad A^\mu_{; \lambda} = A^\mu_{, \lambda} + \Gamma_{\alpha\lambda}^\mu A^\alpha \rightarrow$ 协变微商公式 (推导背面)



$$A^{\mu}_{;\lambda} B_{\mu} + A^{\mu} (B_{\mu,\lambda} - \Gamma^{\sigma}_{\mu\lambda} B_{\sigma}) = A^{\mu}_{,\lambda} B_{\mu} + A^{\mu} B_{\mu,\lambda}$$

$$A^{\mu}_{;\lambda} B_{\mu} + \cancel{A^{\mu} B_{\mu,\lambda}} - \Gamma^{\sigma}_{\mu\lambda} A^{\mu} B_{\sigma} = A^{\mu}_{,\lambda} B_{\mu} + \cancel{A^{\mu} B_{\mu,\lambda}}$$

$$A^{\mu}_{;\lambda} B_{\mu} = A^{\mu}_{,\lambda} B_{\mu} + \Gamma^{\alpha}_{\alpha\lambda} A^{\alpha} B_{\mu}$$

$$\therefore A^{\mu}_{;\lambda} = A^{\mu}_{,\lambda} + \Gamma^{\mu}_{\alpha\lambda} A^{\alpha}$$

是张量场，是张量场的协变导数



(4) 二阶混合张量场 $T^\mu_\nu(x)$

标量为 $T^\mu_\nu A^\nu B_\mu$, 有 $(T^\mu_\nu A^\nu B_\mu)_{;\lambda} = (T^\mu_\nu A^\nu B_\mu)_{;\lambda}$

$$\Rightarrow T^\mu_{\nu;\lambda} = T^\mu_{\nu,\lambda} + \Gamma^\mu_{\rho\lambda} T^\rho_\nu - \Gamma^\rho_{\nu\lambda} T^\mu_\rho$$

类比可以写出

$$\begin{cases} \text{= 阶逆变} & T^\mu_{\nu;\lambda} = T^\mu_{\nu,\lambda} + \Gamma^\mu_{\rho\lambda} T^\rho_\nu - \Gamma^\rho_{\nu\lambda} T^\mu_\rho \\ \text{= 阶协变} & T_{\mu\nu;\lambda} = T_{\mu\nu,\lambda} - \Gamma^\rho_{\mu\lambda} T_{\rho\nu} - \Gamma^\rho_{\nu\lambda} T_{\mu\rho} \end{cases}$$

(5) δ^μ_ν 的协变微商

直接类比 T^μ_ν 给出 $\delta^\mu_{\nu;\lambda} = \delta^\mu_{\nu,\lambda} + \Gamma^\mu_{\rho\lambda} \delta^\rho_\nu - \Gamma^\rho_{\nu\lambda} \delta^\mu_\rho$

δ^μ_ν 分量为常数 故 $\delta^\mu_{\nu,\lambda} = 0$

后两项 $\Gamma^\mu_{\rho\lambda} \delta^\rho_\nu = \Gamma^\mu_{\nu\lambda}$ $\Gamma^\rho_{\nu\lambda} \delta^\mu_\rho = \Gamma^\mu_{\nu\lambda}$ 抵消

从而 $\delta^\mu_{\nu;\lambda} = 0$

结论: δ^μ_ν 的普通微商和协变微商都是零.

(6) 例. $A_{\mu,\nu}$ 的反称组合为张量

$$A_{[\mu,\nu]} = A_{[\mu,\nu]} - \Gamma^\lambda_{[\mu\nu]} A_\lambda \rightarrow \text{挠率张量}$$

左与右 = 为张量, 故 $A_{[\mu,\nu]}$ 为张量

在相中采用对称联络, 即 $\Gamma^\lambda_{[\mu\nu]} = 0$, 有 $A_{[\mu,\nu]} = A_{[\mu,\nu]}$

1.5 测地线方程

普通空间中的直线 $\rightarrow$ 线上任意相邻两点的切向量相互平行的曲线

推广

n 维仿射空间中 $\rightarrow$ 测地线方程

n 维空间中曲线 $x^\mu = x^\mu(\lambda)$ $\mu = 1, 2, \dots, n$; λ 为参量(标量)

任一点切向量 $A^\mu = dx^\mu/d\lambda \rightarrow$ 逆变矢量

定义: 若 P, Q 曲线上相邻两点, 满足 $A^\mu(P \rightarrow Q) \parallel A^\mu(Q)$

那么该曲线为测地线

条件可改写为 $A^\mu(Q) = (1 + f(\lambda) d\lambda) A^\mu(P \rightarrow Q)$

平移公式有 $A^\mu(P \rightarrow Q) = dx^\mu/d\lambda - \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} d\lambda$

切向量微分公式有 $A^\mu(Q) = A^\mu(P) + dA^\mu(P)$
 $= dx^\mu/d\lambda + \frac{d^2 x^\mu}{d\lambda^2} d\lambda$

代入条件有 $\frac{dx^\mu}{d\lambda} + \frac{d^2 x^\mu}{d\lambda^2} d\lambda = [1 + f(\lambda) d\lambda] \left(\frac{dx^\mu}{d\lambda} - \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} d\lambda \right)$
 $= \frac{dx^\mu}{d\lambda} - \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} d\lambda + f(\lambda) \frac{dx^\mu}{d\lambda} d\lambda - 0 d\lambda$
 $\Rightarrow \frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = f(\lambda) \frac{dx^\mu}{d\lambda}$

此乃测地线的微分方程.

若有参量变换 $\lambda = \lambda(\sigma)$, 满足 $\frac{d^2 \sigma}{d\lambda^2} = f(\lambda) \frac{d\sigma}{d\lambda}$

测地线方程简化为 $\frac{d^2 x^\mu}{d\sigma^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\sigma} \frac{dx^\beta}{d\sigma} = 0$

σ 称为仿射参量, 有性质

(a) $f(\sigma) = 0$.

测地线条件 $A^\mu(Q) = A^\mu(P \rightarrow Q)$

其中 $A^\mu \equiv \frac{dx^\mu}{d\sigma}$

(b) 不唯一, 不同的 σ 与 $\tilde{\sigma}$ 满足

$$\frac{d\tilde{\sigma}}{d\sigma} = 0$$

即仿射参量的变换只能是线性变换.



1.6 曲率张量

联络 $\rightarrow$ 决定空间几何性质

↓
非张量 $\rightarrow$ 构成的挠率是张量
构造出曲率张量

考虑协变矢量场 $A_\mu(x) =$ 阶协变微商

$$A_{\lambda;\mu;\nu} = A_{\lambda;\mu;\nu} - \Gamma_{\lambda\nu}^\rho A_{\rho;\mu} - \Gamma_{\mu\nu}^\rho A_{\lambda;\rho}$$

$$= A_{\lambda;\mu;\nu} - \Gamma_{\lambda\mu,\nu}^\rho A_\rho - \Gamma_{\lambda\mu}^\rho A_{\rho;\nu} - \Gamma_{\lambda\nu}^\rho A_{\rho;\mu} + \Gamma_{\lambda\nu}^\rho \Gamma_{\rho\mu}^\sigma A_\sigma - \Gamma_{\mu\nu}^\rho A_{\lambda;\rho}$$

取 $A_{\lambda;\mu;\nu}$ 对 μ 和 ν 的反对称组合

$$A_{\lambda;[\mu;\nu]} = \frac{1}{2} (A_{\lambda;\mu;\nu} - A_{\lambda;\nu;\mu}) = \frac{1}{2} \left(\overset{\text{曲}}{R_{\lambda\mu\nu}^\rho} A_\rho - 2 \overset{\text{挠}}{[\Gamma_{\mu\nu}^\rho] A_{\lambda;\rho}} \right) \quad \text{当曲率挠率均零, } A_\mu \text{ 两次协变微商可以换序}$$

其中 $R_{\lambda\mu\nu}^\rho = -\Gamma_{\lambda\mu,\nu}^\rho + \Gamma_{\lambda\nu,\mu}^\rho - \Gamma_{\lambda\mu}^\sigma \Gamma_{\sigma\nu}^\rho + \Gamma_{\lambda\nu}^\sigma \Gamma_{\sigma\mu}^\rho \rightarrow$ 定义式 for 曲率 (1,3) 阶张量

↓
性质: (1) 曲率完全由联络及其一阶微商决定

$$(2) R_{\lambda\mu\nu}^\rho = -R_{\lambda\nu\mu}^\rho$$

内若曲率与挠率恒为零, 总有一个适当的变换 $x^\mu \rightarrow \tilde{x}^\mu$

使得 $\Gamma_{\mu\nu}^\lambda = 0$ (V内)

1) 于是由矢量平移公式有 $\delta A_\mu = 0 \rightarrow$ V内矢量的平移不改变其分量

2) 测地线方程简化为 $\frac{d^2 x^\mu}{d\sigma^2} = 0 \rightarrow$ V内测地线是直线

曲/挠为零



$\rightarrow$ V内空间是“平坦的” $\rightarrow$ 平移增量路径无关
 $\rightarrow$ 闭合路径一圈不变/增量为零

两种独立缩并方式 $\left\{ \begin{array}{l} A_{\mu\nu} \equiv R_{\lambda\mu\nu}^\lambda \\ R_{\mu\nu} \equiv R_{\mu\nu\lambda}^\lambda \end{array} \right.$



二、黎曼 (Riemann) 几何

0

仿射空间中引入度规场和不变距离就构成了黎曼空间。

2.1 黎曼空间和度规张量

二次型 $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ 定义两点距离

令 $g_{\mu\nu} = g_{\nu\mu}$ 得到 Riemann Space.

四维闵氏空间, 取 $x^0 = ct, x^1 = x, x^2 = y, x^3 = z$

不变距离公式 $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$

$$g_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} = g_{\nu\mu} = \eta_{\mu\nu}$$

若 $g_{\mu\nu} = \begin{cases} +1 \text{ 或 } -1 & \mu=\nu \\ 0 & \mu \neq \nu \end{cases}$ = "平坦的"

2.2 张量指标的升降

黎曼 RS 中, 可以有 $T_\mu = g_{\mu\nu} T^\nu$ 与 $T_{\mu\nu} = g_{\mu\alpha} T^\alpha_\nu$, 这是降指标

$\det|g_{\mu\nu}| \neq 0$, 可以定义 $g^{\mu\nu}$, 满足 $g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu_\nu$

用于升指标 $\begin{cases} T^\mu = g^{\mu\nu} T_\nu \\ T^{\mu\nu} = g^{\mu\alpha} T^\nu_\alpha \end{cases}$

eq1. 矢量模方 $g_{\mu\nu} T^\mu T^\nu = T_\nu T^\nu = g^{\mu\nu} T_\mu T_\nu$

eq2. 不变距离 $ds^2 = g_{\mu\nu} dx^\mu dx^\nu = dx_\nu dx^\nu = g^{\mu\nu} dx_\mu dx_\nu$



2.3 克里斯朵夫联络 (Christoffel)

②

P点逆变矢量: $A^\mu(P) \xrightarrow{\Gamma_{\lambda\mu}^\alpha} \text{邻点Q: } A^\mu(P \rightarrow Q)$

$$\text{有 } A^\mu(P \rightarrow Q) = A^\mu(P) - \Gamma_{\lambda\mu}^\alpha A^\lambda(P) dx^\lambda$$

要求: 平移不改变矢量长度,

$$\text{那么 } g_{\mu\nu}(Q) A^\mu(P \rightarrow Q) A^\nu(P \rightarrow Q) = g_{\mu\nu}(P) A^\mu(P) A^\nu(P)$$

$$\text{度规场微分公式 } g_{\mu\nu}(Q) = g_{\mu\nu}(P) + g_{\mu\nu,\lambda}(P) dx^\lambda$$

代入整理有 $g_{\mu\nu,\lambda} - g_{\alpha\nu} \Gamma_{\mu\lambda}^\alpha - g_{\mu\alpha} \Gamma_{\nu\lambda}^\alpha = 0$ ① → 保持长度的联络满足的方程

指标循环替换 $\mu \rightarrow \nu \rightarrow \lambda \rightarrow \mu$,

$$\begin{cases} g_{\mu\lambda,\mu} - g_{\alpha\lambda} \Gamma_{\nu\mu}^\alpha - g_{\nu\alpha} \Gamma_{\lambda\mu}^\alpha = 0 & \text{②} \\ g_{\lambda\mu,\nu} - g_{\mu\alpha} \Gamma_{\lambda\nu}^\alpha - g_{\lambda\alpha} \Gamma_{\mu\nu}^\alpha = 0 & \text{③} \end{cases}$$

同时①由协变微商定义可写为

$$g_{\mu\nu;\lambda} = 0 \text{ 或 } g^{\mu\nu}_{;\lambda} = 0$$

说明了度规张量在协变微商下行为和常数张量普通微商下一致。

重要定理: x^μ 下 P 点联络为 $(\Gamma_{\mu\nu}^\lambda)_P$, 总可找到 $\tilde{x}^\mu \rightarrow \tilde{x}^\mu$, 使得 $(\tilde{\Gamma}_{\mu\nu}^\lambda)_P = 0$

不妨令 $\tilde{x}^\mu \rightarrow \tilde{x}^\mu$ 满足: $x^\mu - x^\mu_P = \tilde{x}^\mu + \frac{1}{2} C_{\alpha\beta}^\mu \tilde{x}^\alpha \tilde{x}^\beta$ 其中 $C_{\alpha\beta}^\mu = C_{\beta\alpha}^\mu$

$$\text{P点旧坐标 } x^\mu_P, \text{ 新坐标 } \tilde{x}^\mu_P = 0 \quad \text{且} \quad \left(\frac{\partial x^\mu}{\partial \tilde{x}^\lambda} \right)_P = \delta_\lambda^\mu, \quad \left(\frac{\partial \tilde{x}^\mu}{\partial x^\lambda} \right)_P = \delta_\lambda^\mu, \quad \left(\frac{\partial^2 x^\mu}{\partial \tilde{x}^\lambda \partial \tilde{x}^\nu} \right)_P = C_{\lambda\nu}^\mu$$

$$\text{度联络变换 } \tilde{\Gamma}_{\mu\nu}^\lambda = (\Gamma_{\mu\nu}^\lambda)_P + C_{\mu\nu}^\lambda, \text{ 只要令 } C_{\mu\nu}^\lambda = -(\Gamma_{\mu\nu}^\lambda)_P \text{ 即可.}$$

含义: x^μ 下 P 点联络为零, 那么 $(g_{\mu\nu,\lambda})_P = 0$, 即 P 点附近 dx^μ 为微小量可略区域内 $g_{\mu\nu}$ 近似为常数



$$[1] \quad g_{\mu\nu}(\partial) A^\mu(P) A^\nu(P) = g_{\mu\nu}(P) A^\mu(P) A^\nu(P)$$

$$[g_{\mu\nu}(P) + g_{\mu\nu,\lambda}(P) d\lambda] [\cancel{A^\mu(P)} A^\mu(P) - \Gamma_{\nu\lambda}^\mu(P) A^\nu(P) d\lambda] [A^\nu(P) - \Gamma_{\alpha\lambda}^\nu(P) A^\alpha(P) d\lambda] = g_{\mu\nu}(P) A^\mu(P) A^\nu(P).$$

$$\cancel{g_{\mu\nu}(P) A^\mu(P) A^\nu(P)} + g_{\mu\nu}(P) [-A^\mu(P) \Gamma_{\alpha\lambda}^\nu(P) A^\alpha(P) - A^\nu(P) \Gamma_{\alpha\lambda}^\mu(P) A^\alpha(P)] d\lambda \quad \textcircled{1} \nu \rightarrow \alpha \rightarrow \nu \quad \textcircled{2} \nu \rightarrow \mu \rightarrow \alpha.$$

$$+ g_{\mu\nu,\lambda}(P) A^\mu(P) A^\nu(P) d\lambda = \cancel{g_{\mu\nu}(P) A^\mu(P) A^\nu(P)}.$$

$$- g_{\mu\alpha} A^\mu(P) \Gamma_{\nu\lambda}^\alpha(P) A^\nu(P) - g_{\mu\alpha} A^\mu(P) \Gamma_{\nu\lambda}^\alpha(P) A^\nu(P) + g_{\mu\nu,\lambda}(P) A^\mu(P) A^\nu(P) d\lambda = 0$$

$$[2] \quad (g_{\mu\nu,\lambda} - g_{\alpha\nu} \Gamma_{\mu\lambda}^\alpha - \cancel{g_{\mu\alpha} \Gamma_{\nu\lambda}^\alpha} = 0)$$

$$+ (g_{\nu\lambda,\mu} - \cancel{g_{\alpha\mu} \Gamma_{\nu\lambda}^\alpha} - g_{\mu\alpha} \Gamma_{\nu\lambda}^\alpha = 0)$$

$$- (g_{\lambda\mu,\nu} - \cancel{g_{\alpha\mu} \Gamma_{\lambda\nu}^\alpha} - \cancel{g_{\mu\alpha} \Gamma_{\lambda\nu}^\alpha} = 0)$$

$$\Rightarrow g_{\mu\nu,\lambda} + g_{\nu\lambda,\mu} - g_{\lambda\mu,\nu} = g_{\alpha\nu} \Gamma_{\mu\lambda}^\alpha + g_{\mu\alpha} \Gamma_{\nu\lambda}^\alpha$$

$$= 2 g_{\alpha\nu} \Gamma_{\lambda\mu}^\alpha$$



2.4 黎曼空间中的测地线

任一曲线, 可引入一个标量积分

$$s = \int_{P_0}^P ds, \quad s \text{ 为 } P_0 \rightarrow P \text{ 固有长度.}$$

以 s 为参量, 记 u^μ 为

$$u^\mu = dx^\mu/ds \quad \text{且有} \quad g_{\mu\nu} u^\mu u^\nu = 1$$

↓ 切变微分

$$g_{\mu\nu} u^\mu_{;\lambda} u^\nu + g_{\mu\nu} u^\mu u^\nu_{;\lambda} = 0$$

$$\Rightarrow u_\nu u^\nu_{;\lambda} = 0$$

测地线方程为

$$\frac{du^\mu}{ds} + \Gamma_{\rho\sigma}^\mu u^\rho u^\sigma = f(s) u^\mu$$

$$\Rightarrow f(s) = u_\mu \left(\frac{du^\mu}{ds} + \Gamma_{\rho\sigma}^\mu u^\rho u^\sigma \right) \quad \text{注意到} \quad \frac{du^\mu}{ds} = u^\mu_{;\tau} u^\tau$$

$$\Rightarrow f(s) = u_\mu (u^\mu_{;\tau} u^\tau + \Gamma_{\rho\sigma}^\mu u^\rho u^\sigma) = u_\mu u^\mu_{;\tau} u^\tau = 0$$

$$\Rightarrow s \text{ 是仿射参量, 测地线方程 } \frac{du^\mu}{ds} + \Gamma_{\rho\sigma}^\mu u^\rho u^\sigma = 0$$

↓
重要性质: 使泛函 $L = \int_A^B ds$ 取极值.



2.5 黎曼空间的曲率张量

联络是对称的 $\Rightarrow \begin{cases} R_{\lambda\mu\nu}^{\rho} = -R_{\lambda\nu\mu}^{\rho} \\ R_{\lambda\mu\nu}^{\rho} + R_{\mu\nu\lambda}^{\rho} + R_{\nu\lambda\mu}^{\rho} = 0 \end{cases}$

引用(10.4)阶 $R_{\rho\lambda\mu\nu} = g_{\rho\sigma} R_{\lambda\mu\nu}^{\sigma}$, $R_{\rho\lambda\mu\nu}$ 有四组对称性

① 对 μ 和 ν 反对称 $R_{\rho\lambda\mu\nu} = -R_{\rho\lambda\nu\mu}$

② 对 ρ 和 λ 反对称 $R_{\rho\lambda\mu\nu} = -R_{\lambda\rho\mu\nu}$

③ 对 $\rho\lambda$ 和 $\mu\nu$ 对称 $R_{\rho\lambda\mu\nu} = +R_{\mu\nu\rho\lambda}$

④ 对 $\lambda\mu\nu$ 反对称 $R_{\rho\lambda\mu\nu} + R_{\rho\mu\nu\lambda} + R_{\rho\nu\lambda\mu} = 0$

$\Rightarrow$ 四维RS的 $R_{\rho\lambda\mu\nu}$ 共有20个独立分量.

唯一非零的缩并 $R_{\mu\nu} = R_{\mu\nu\lambda}^{\lambda} \rightarrow$ Ricci 张量

$$R_{\mu\nu} = g^{\lambda\alpha} R_{\alpha\mu\nu\lambda} = g^{\lambda\alpha} R_{\nu\lambda\alpha\mu} = g^{\lambda\alpha} R_{\lambda\nu\mu\alpha}$$

$\Rightarrow R_{\mu\nu} = R_{\nu\mu}$ Ricci 张量是对称的

再上升一个指标并且缩并 $R \equiv g^{\mu\nu} R_{\mu\nu} = R_{\mu}^{\mu}$ 称为标量曲率

$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \rightarrow$ 爱因斯坦张量 (对称)

2.6 毕安基 (Bianchi) 恒等式

RT 定义为: $R_{\lambda\mu\nu}^{\rho} = -\Gamma_{\lambda\mu,\nu}^{\rho} + \Gamma_{\lambda\nu,\mu}^{\rho} - \Gamma_{\lambda\mu}^{\sigma}\Gamma_{\sigma\nu}^{\rho} + \Gamma_{\lambda\nu}^{\sigma}\Gamma_{\sigma\mu}^{\rho}$

一阶协变微商 $R_{\lambda\mu\nu;\tau}^{\rho} + R_{\lambda\nu\tau;\mu}^{\rho} + R_{\lambda\tau\mu;\nu}^{\rho} = 0$ Bianchi 恒等式

对 ρ, τ 缩并有

$$R_{\lambda\mu\nu;\tau}^{\tau} + R_{\lambda\nu;\mu} - R_{\lambda\mu;\nu} = 0$$

乘 $g^{\mu\lambda}$ 再缩并 ($g^{\mu\lambda}_{;\lambda} = 0$)

$$(g^{\mu\lambda} R_{\lambda\mu\nu;\tau}^{\tau}) + (g^{\mu\lambda} R_{\lambda\nu})_{;\mu} - (g^{\mu\lambda} R_{\lambda\mu})_{;\nu} = 0$$

$$\Rightarrow (R_{\mu}^{\nu} - \frac{1}{2} \delta_{\mu}^{\nu} R)_{;\nu} = 0$$

爱因斯坦张量的协变散度恒为零



2.7 李(Lie)微商

$x^\mu \rightarrow x^\mu(x')$ 坐标变换/同一空间内的映射

无穷小映射 $x^\mu = x^\mu + \varepsilon \xi^\mu \rightarrow$ 生成元, 完全决定了一个无穷小映射

张量场 $T(x)$, P 与 Q 为邻近点, $T(P)$ 移至 Q 点为 $T(P \rightarrow Q)$

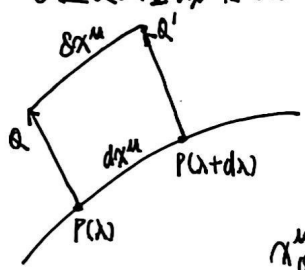
$$\mathcal{L}_\xi T(x) = \lim_{\varepsilon \rightarrow 0} \frac{T(Q) - T(P \rightarrow Q)}{\varepsilon}$$

① 标量场 $\varphi(x)$

定义 $\varphi(P \rightarrow Q) = \varphi(P)$

$$\begin{aligned} \mathcal{L}_\xi \varphi(x) &= \lim_{\varepsilon \rightarrow 0} \frac{\varphi(Q) - \varphi(P \rightarrow Q)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{\varphi(Q) - \varphi(P)}{\varepsilon} \\ &= \lim_{\varepsilon \rightarrow 0} \varphi_{, \mu} dx^\mu / \varepsilon = \varphi_{, \mu} \xi^\mu \end{aligned}$$

② 逆变张量场 $k^\mu(x)$



$$k^\mu(P) = \left(\frac{dx^\mu}{d\lambda} \right)_P$$

$$\text{令 } \overline{P}^\mu = (dx^\mu)_P$$

$$\left\{ \begin{array}{c} P \\ P' \end{array} \right\} \xrightarrow{\text{映射}} \left\{ \begin{array}{c} Q \\ Q' \end{array} \right\} \quad \overline{Q}^\mu \text{ 代表 } k^\mu(P \rightarrow Q)$$

$$x^\mu_Q = x^\mu + \varepsilon \xi^\mu(x)$$

$$x^\mu_{Q'} = x^\mu + dx^\mu + \varepsilon \xi^\mu(x + dx)$$

$$\delta x^\mu = \overline{Q}^\mu = x^\mu + dx^\mu + \varepsilon \xi^\mu(x + dx) - x^\mu - \varepsilon \xi^\mu(x)$$

$$= dx^\mu + \varepsilon \xi^\mu_{, \nu} dx^\nu + O(\dots)$$

$$k^\mu(P \rightarrow Q) = \frac{\delta x^\mu}{d\lambda} = \frac{dx^\mu}{d\lambda} + \varepsilon \xi^\mu_{, \nu} \frac{dx^\nu}{d\lambda}$$

$$= k^\mu(P) + \varepsilon \xi^\mu_{, \nu}(P) k^\nu(P)$$

$$\mathcal{L}_\xi k^\mu(x) = \lim_{\varepsilon \rightarrow 0} \frac{k^\mu(Q) - k^\mu(P \rightarrow Q)}{\varepsilon}$$

$$= k^\mu_{, \nu} \xi^\nu - \xi^\mu_{, \nu} k^\nu$$

③ 协变张量场 $p_\mu(x)$

$$\mathcal{L}_\xi (p_\mu k^\mu) = (\mathcal{L}_\xi p_\mu) k^\mu + p_\mu (\mathcal{L}_\xi k^\mu)$$

$$\text{而又 } \mathcal{L}_\xi (p_\mu k^\mu) = (p_\mu k^\mu)_{, \sigma} \xi^\sigma$$

$$= p_{\mu, \sigma} k^\mu \xi^\sigma + p_\mu k^\mu_{, \sigma} \xi^\sigma$$

$$= \mathcal{L}_\xi p_\mu = p_{\mu, \sigma} \xi^\sigma + \xi^\sigma_{, \mu} p_\sigma$$

李微商 取决于 $\left\{ \begin{array}{l} \text{被微商的张量场} \\ \text{映射的生成元} \end{array} \right.$
(不需先有联络)

规定联络, 李微商可用协变微商表示

2.8 等度规映射和凯林(Killing)矢量场

P, Q 为相邻点, 坐标差 dx^μ

不变距离平方为 $ds_{PQ}^2 = g_{\mu\nu}(P) dx^\mu dx^\nu$

无穷小映射

P', Q' , 坐标差 δx^μ

不变距离平方为 $ds_{P'Q'}^2 = g_{\mu\nu}(P) \delta x^\mu \delta x^\nu$

若对 $\forall P, Q$ 都有映射前后不变距离相等,

即 $ds_{PQ}^2 = ds_{P'Q'}^2$

那么这种映射为 等度规映射。

生成元 $\xi^\mu(x)$ 为凯林矢量场 $\rightarrow$ 刻画时空结构的几何对称性质

由于 P 与 P' 是映射的对应点, 坐标差 $\varepsilon \xi^\mu$, 所以

$$g_{\mu\nu}(P') = g_{\mu\nu}(P) + g_{\mu\nu,\lambda}(P) \varepsilon \xi^\lambda$$

$P'Q'$ 是 PQ 的映射, 那么

$$\delta x^\mu = dx^\mu + \varepsilon \xi^\mu_{,\lambda} dx^\lambda$$

那么不变距离相等会导出

$$g_{\mu\nu,\lambda} \xi^\lambda + g_{\lambda\nu,\mu} \xi^\lambda + g_{\mu\lambda} \xi^\lambda_{,\nu} = 0$$

$$\text{等价于 } \xi_{\mu;\nu} + \xi_{\nu;\mu} = 0$$

$$\text{也可表示为 } \xi_{\mu;\nu} + \xi_{\nu;\mu} = 0$$

凯林方程

RS 具有 时间平移不变性

某方向空间平移不变性

$\rightarrow$ 凯林场

沿时间轴的平移矢量

该方向的平移矢量



$$ds^2_P = g_{\mu\nu}(P) dx^\mu dx^\nu = ds^2_{P'} = g_{\mu\nu}(P') \delta x^\mu \delta x^\nu$$

$$g_{\mu\nu}(P) dx^\mu dx^\nu = [g_{\mu\nu}(P) + g_{\mu\nu,\lambda}(P) \varepsilon^\lambda] [dx^\mu + \varepsilon^\mu_{,\lambda} dx^\lambda] [dx^\nu + \varepsilon^\nu_{,\lambda} dx^\lambda]$$

$$\begin{aligned} \cancel{g_{\mu\nu}(P) dx^\mu dx^\nu} &= \cancel{g_{\mu\nu}(P) dx^\mu dx^\nu} + g_{\mu\nu,\lambda}(P) \varepsilon^\lambda dx^\mu dx^\nu \quad ① \\ &+ g_{\mu\nu}(P) dx^\mu \varepsilon^\nu_{,\lambda} dx^\lambda \quad ② + g_{\mu\nu,\lambda}(P) \varepsilon^\lambda dx^\mu \varepsilon^\nu_{,\lambda} dx^\lambda \quad x. \\ &+ g_{\mu\nu}(P) \varepsilon^\mu_{,\lambda} dx^\lambda dx^\nu \quad ③ + g_{\mu\nu,\lambda}(P) \varepsilon^\lambda \varepsilon^\mu_{,\lambda} dx^\lambda dx^\nu \quad x. \\ &+ g_{\mu\nu}(P) \varepsilon^\mu_{,\lambda} dx^\lambda \varepsilon^\nu_{,\lambda} dx^\lambda \quad x. + g_{\mu\nu,\lambda}(P) \varepsilon^\lambda \varepsilon^\mu_{,\lambda} dx^\lambda \varepsilon^\nu_{,\lambda} dx^\lambda \quad x. \end{aligned}$$

$$① = g_{\mu\nu,\lambda} \varepsilon^\lambda \underline{dx^\mu dx^\nu}$$

$$v \leftrightarrow \lambda \quad ② = g_{\mu\lambda} \varepsilon^\lambda_{,\nu} \underline{dx^\mu dx^\nu}$$

$$u \leftrightarrow \lambda \quad ③ = g_{\lambda\nu} \varepsilon^\lambda_{,\mu} \underline{dx^\mu dx^\nu}$$

$$? \quad g_{\mu\nu,\lambda} \varepsilon^\lambda + g_{\mu\lambda} \varepsilon^\lambda_{,\nu} + g_{\lambda\nu} \varepsilon^\lambda_{,\mu} = 0$$



三、相对论性的引力理论

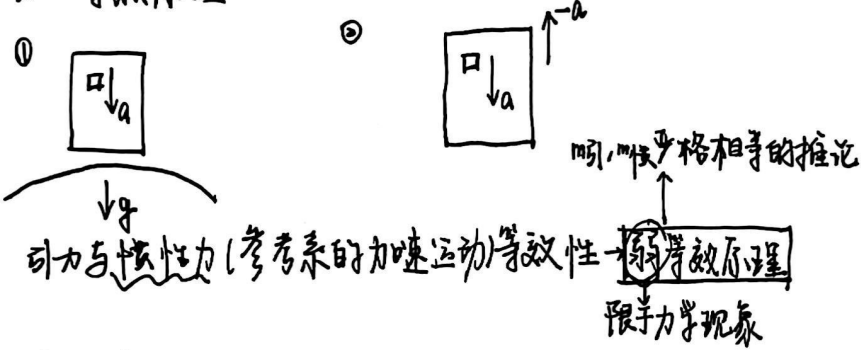
爱因斯坦 $\rightarrow$ 广义相对论
(协变的引力理论) $\left\{ \begin{array}{l} \text{等效原理} \\ \text{引力场方程} \end{array} \right.$

3.1 引力质量与惯性质量的等同性

牛顿 $\left\{ \begin{array}{l} \text{动力学 } F = ma \quad \text{惯性质量} \\ \text{引力 } F = \frac{GMm}{r^2} \quad \text{引力质量} \end{array} \right.$

量值上, $m_{\text{引}} = m_{\text{惯}}$ 是一个观测事实

3.2 等效原理



$\downarrow$ 进一步
任何物理实验不能区分引力或惯性力的效果 $\rightarrow$ 强等效原理

$\downarrow$
“引力和惯性力物理效果上完全没区别”
或: 局域参考系的加速及有相对的意义”

3.3 引力几何化. ($c=1$ 自然单位制)

质点相对闵氏空间中惯性系自由运动

动力学方程 $d^2 X^\mu / ds^2 = 0$ (也是测地线方程)

$X^\mu = (T, X, Y, Z)$ 闵氏坐标

非惯性系 x^μ

广义坐标变换 $x^\mu = x^\mu(x)$

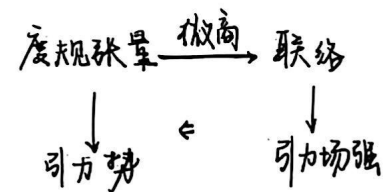
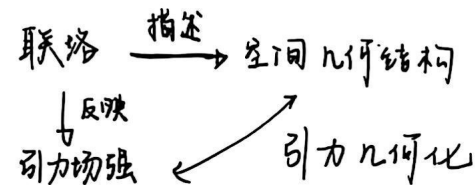
反变换 $X^\mu = X^\mu(x)$

→ 动力学方程 $\frac{d^2 x^\mu}{ds^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds} = 0$ 非惯性系中自由粒子动力学方程

其中 $\Gamma_{\alpha\beta}^\mu = \frac{\partial^2 X^\mu}{\partial x^\alpha \partial x^\beta} \frac{\partial x^\nu}{\partial X^\mu}$

第一项 粒子加速度

第二项 单位质量粒子所受引力



3.4 弱引力场中的自由粒子

动力学方程 $\frac{d^2 x^\mu}{ds^2} = -\Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds}$

(a) 引力场是弱场

令 $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, 有 $|h_{\mu\nu}| \ll 1$

$\hookrightarrow$ 闵氏度规

(b) 引力场静态

$g_{\mu\nu,0} = h_{\mu\nu,0} = 0$

(c) 引力场空间缓变

$g_{\mu\nu,i} = h_{\mu\nu,i} \ll 1$ ($i=1,2,3$)

(d) 粒子运动低速

$\frac{dx^i}{dx^0} \ll 1$

→ 测地线方程还原为牛顿方程

$$\frac{d^2 x^i}{dt^2} = -\frac{\partial \varphi}{\partial x^i}$$

$t \equiv x^0$, φ 为牛顿引力势

证明见反面.



前3个条件关系给出 $\Gamma_{\alpha\beta}^{\mu}$ 为小量,

保留至一级小量

$$\Gamma_{\alpha\beta}^{\mu} = \frac{1}{2} \eta^{\mu\rho} (h_{\rho\alpha,\beta} + h_{\rho\beta,\alpha} - h_{\alpha\beta,\rho})$$

利用条件4

$$\begin{cases} \frac{d^2 x^0}{ds^2} = 0 & \textcircled{1} \\ \frac{d^2 x^i}{ds^2} = -\Gamma_{00}^i \left(\frac{dx^0}{ds} \right)^2 & \textcircled{2} \end{cases}$$

①解出 $x^0 = \tau + \text{const}$

那么②变为 $\frac{d^2 x^i}{dt^2} = -\frac{1}{2} h_{00,i} \rightarrow$ 牛顿方程的形式

有关系 $\varphi = -\frac{1}{2} h_{00} + \text{const}$

∞ 处, $\varphi=0$ 处有 $h_{00}=0$, 那么 $\varphi = -\frac{1}{2} h_{00}$

或 $g_{00} = -1 + h_{00} = -1 - 2\varphi$

∴ 引力场中的牛顿方程仅对静态、缓变、弱引力场中的自由粒子适用。



3.5 爱因斯坦引力场方程

牛顿引力场势的分布取决于静态物质密度分布
+ 张量理论中物质密度是动量能量张量的分量

↓ 定规场取决于物质的动量能量张量

场方程 $F_{\mu\nu} = T_{\mu\nu}$

牛顿引力方程 $\nabla^2 \varphi = 4\pi G \rho$

→ $F_{\mu\nu}$ 最一般为 $F_{\mu\nu} = \alpha R_{\mu\nu} + \beta g_{\mu\nu} R + \gamma g_{\mu\nu}$

能量动量守恒 → $T^{\mu}_{\nu;\mu} = 0 = F^{\mu}_{\nu;\mu}$

+ $\beta = -\frac{\alpha}{2}$ 即把 $R_{\mu\nu}$ 和 R 组成 $G_{\mu\nu}$

于是场方程 $G_{\mu\nu} + \lambda g_{\mu\nu} = -\kappa T_{\mu\nu}$

最简单的情况 $\lambda=0$

此时 $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = -\kappa T_{\mu\nu}$ 指标上升后缩并 → $R = \kappa T \equiv T^{\mu}_{\mu}$ 为动量能量张量的迹

这是爱因斯坦引力场方程

代回, 有一等价表述 $R_{\mu\nu} = -\kappa (T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T)$

$T_{\mu\nu} = 0$ 时, $R_{\mu\nu} = 0$ 为真空区场方程



3.6 场方程的牛顿近似

考虑由“非相对性理想流体”构成的介质

动-能张量 $T^{\mu\nu} = \rho u^\mu u^\nu$ $\left\{ \begin{array}{l} \rho \text{ 密度} \\ u^\mu = \frac{dx^\mu}{d\tau} \end{array} \right.$ 介质四维速

$d\tau^2 = -ds^2 = -g_{\mu\nu} dx^\mu dx^\nu$ $\leftarrow d\tau$ 为固有时

u^μ 归一为 $g_{\mu\nu} u^\mu u^\nu = -1$

$(= g_{\mu\nu} \frac{dx^\mu dx^\nu}{d\tau^2} = g_{\mu\nu} \frac{dx^\mu dx^\nu}{-g_{\mu\nu} dx^\mu dx^\nu})$

用相对介质静止的系, u^μ 形式 $u^\mu = \frac{1}{\sqrt{-g_{00}}} (1, 0, 0, 0)$

$\Rightarrow$ 动-能张量唯一非零分量 $T^{00} = -\rho/g_{00}$

之后计算①静态②缓变③弱场近似下的 Ricci Tensor $R_{\mu\nu}$.

$R_{\mu\nu} = R^\lambda_{\mu\nu\lambda}$
 $= \Gamma^\lambda_{\mu\lambda,\nu} - \Gamma^\lambda_{\mu\nu,\lambda} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\rho} - \Gamma^\lambda_{\mu\nu} \Gamma^\rho_{\lambda\rho}$

保留至一级小量, 有联络

$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\rho} (h_{\rho\mu,\nu} + h_{\rho\nu,\mu} - h_{\mu\nu,\rho})$

即 $\left\{ \begin{array}{l} \Gamma^0_{00} = 0 \\ \Gamma^0_{0i} = -\frac{1}{2} h_{00,i} \\ \Gamma^0_{ij} = -\frac{1}{2} (h_{0i,j} + h_{0j,i}) \end{array} \right.$

与 $\left\{ \begin{array}{l} \Gamma^i_{00} = -\frac{1}{2} h_{00,i} \\ \Gamma^i_{0j} = \frac{1}{2} (h_{0i,j} - h_{0j,i}) \\ \Gamma^i_{jk} = \frac{1}{2} (h_{ij,k} + h_{ik,j} - h_{jk,i}) \end{array} \right.$

忽略联络二阶项, Ricci Tensor 为

$R_{\mu\nu} = \Gamma^\lambda_{\mu\lambda,\nu} - \Gamma^\lambda_{\mu\nu,\lambda}$

那么 $R_{00} = \frac{1}{2} h_{00,i,i}$

$R_{0i} = \frac{1}{2} (h_{k0,i,k} - h_{0i,k,k})$

$R_{ij} = \frac{1}{2} (h_{kk,i,j} - h_{ki,j,k} - h_{kj,i,k} + h_{ij,k,k})$

换为逆变 $R^{\mu\nu} = g^{\mu\alpha} g^{\nu\beta} R_{\alpha\beta} = g^{\mu\alpha} g^{\nu\beta} R_{\alpha\beta}$

一级近似下, $\left\{ \begin{array}{l} R^{00} = R_{00} \\ R^{0i} = -R_{0i} \\ R^{ij} = R_{ij} \end{array} \right.$

00分量下, 场方程

$R^{00} = -\kappa (T^{00} - \frac{1}{2} g^{00} T)$

$\Rightarrow h_{00,i,i} = -\kappa \rho$

牛顿引力势 $\varphi = -\frac{1}{2} h_{00}$

于是 $\nabla^2 \varphi = \frac{1}{2} \kappa \rho = 4\pi G \rho$

$\Rightarrow \kappa = 8\pi G$

(一般单位制下 $\kappa = 8\pi G/c^4$)



3.7 谐和坐标条件

$$G_{\mu\nu} = -8\pi G T_{\mu\nu}$$

10个方程 $\xrightarrow{G_{\mu\nu;\mu} = 0}$ 6个独立

$g_{\mu\nu}$ 有4个不能为场方程所确定的自由度

常采用谐和坐标条件为限制条件,

要求坐标选择满足 $\square^\lambda \equiv g^{\mu\nu} \Gamma_{\mu\nu}^\lambda = 0$

$$\text{或 } g^{\mu\nu} g^{\rho\sigma} (g_{\rho\mu,\nu} + g_{\rho\nu,\mu} - g_{\mu\nu,\rho}) = 0$$

应用: 弱引力场下线性近似理论

$$\text{度规 } g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, |h_{\mu\nu}| \ll 1$$

只保留 $h_{\mu\nu}$ 线性项.

$$\text{有 } \Gamma_{\alpha\beta}^\mu = \frac{1}{2} g^{\mu\nu} (h_{\alpha\nu,\beta} + h_{\beta\nu,\alpha} - h_{\alpha\beta,\nu})$$

$$= \frac{1}{2} (h_{\alpha,\beta}^\mu + h_{\beta,\alpha}^\mu - h_{\alpha\beta}^{\prime\mu})$$

$$R$$
 为 $R_{\mu\nu} = \Gamma_{\mu\lambda,\nu}^\lambda - \Gamma_{\mu\nu,\lambda}^\lambda$

$$\equiv \frac{1}{2} (h_{\mu\nu,\alpha}^{\prime\alpha} + h_{\mu,\nu}^{\prime\alpha} - h_{\mu,\nu,\alpha}^{\prime\alpha} - h_{\nu,\mu,\alpha}^{\prime\alpha})$$

$$\text{其中 } h \equiv h_{\alpha}^{\alpha} = \eta^{\alpha\beta} h_{\alpha\beta}$$

$$\text{定义 } \bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$$

$$\text{线性化场方程 } \bar{R}_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} R = -8\pi G T_{\mu\nu}$$

$$\text{即 } \bar{h}_{\mu\nu,\alpha}^{\prime\alpha} + \eta_{\mu\nu} \bar{h}_{\alpha\beta}^{\prime\alpha\beta} - \bar{h}_{\mu\alpha,\nu}^{\prime\alpha} - \bar{h}_{\nu\alpha,\mu}^{\prime\alpha} = -16\pi G T_{\mu\nu}$$

谐和条件线性化 $\bar{h}_{\mu\alpha}^{\prime\alpha} = 0$
(引力势洛伦兹规范条件)

$$\text{场方程为 } \bar{h}_{\mu\nu,\alpha}^{\prime\alpha} = 16\pi G T_{\mu\nu}$$

$$\text{度规场 } g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} = \eta_{\mu\nu} + \bar{h}_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \bar{h}$$

ϕ 是谐和量 $\rightarrow$ 满足谐和方程 $\square\phi \equiv (g^{\mu\nu} \phi_{;\mu})_{;\nu} = 0$

$$\square\phi = g^{\mu\nu} \frac{\partial^2 \phi}{\partial x^\mu \partial x^\nu} - \Gamma^\lambda_{\mu\nu} \frac{\partial \phi}{\partial x^\lambda}$$

取 ϕ 为坐标 x^μ

$$\frac{\partial^2 x^\mu}{\partial x^\lambda \partial x^\rho} = 0 \rightarrow \square x^\mu = -\Gamma^\lambda_{\mu\nu} \frac{\partial x^\mu}{\partial x^\lambda} = -\Gamma^\mu_{\mu\nu}$$

若坐标选取满足 $\Gamma^\mu_{\mu\nu} = 0$, 便有 $\square x^\mu = 0$

3.8 引力波

$\bar{h}_{\mu\nu,\alpha}^{\prime\alpha} = 16\pi G T_{\mu\nu}$ 是真空中波动方程

真空中, $T_{\mu\nu} = 0$, 那么 $R_{\mu\nu} = 0$

波动方程为 $\bar{h}_{\mu\nu,\alpha}^{\prime\alpha} = 0$

洛伦兹规范 $\bar{h}_{\mu\alpha}^{\prime\alpha} - \frac{1}{2} \eta_{\mu\alpha} \bar{h}^{\prime\alpha} = 0$

考虑 x 向单色平面波 $h_{\mu\nu} = A_{\mu\nu} e^{i(kx - \omega t)}$

$h_{\mu\nu,\alpha}^{\prime\alpha}$ 决定了 $k = \omega$

要求振幅满足 $\bar{A}_\mu^{\prime\alpha} = \bar{A}_\mu^0$ 其中 $\bar{A}_\mu^{\prime\alpha} = A_\mu^{\prime\alpha} - \frac{1}{2} \delta_\mu^\alpha A^{\prime\alpha}$

采用 $x^\mu \rightarrow x^\mu + \xi^\mu$ 中 $\xi^\mu = \xi_\mu^{(0)} e^{i(kx - \omega t)}$

适当选取 $\xi_\mu^{(0)}$ 可以有

$$\bar{A}_1^0 = \bar{A}_2^0 = \bar{A}_3^0 = 0$$

$$\bar{A}_2^1 + \bar{A}_3^2 = 0$$

沿 $x^1 = x$ 方向传播 $\rightarrow$ 横波规范

余下非零振幅 $\begin{pmatrix} \bar{A}_2^2 & \bar{A}_2^3 \\ \bar{A}_3^2 & \bar{A}_3^3 \end{pmatrix}$ 对称且无迹, 仅2个独立分量.

由于 $\bar{A}_\alpha^{\prime\alpha} = 0$, 有 $A_{\mu\nu} = \bar{A}_{\mu\nu}$

沿 $x^1 = x$ 方向平面引力波的振幅 $A_{\mu\nu}$

在横向无迹洛伦兹规范下仅2个独立分量,

$$\text{可取为 } \begin{cases} A_{22} \\ A_{22} - A_{33} \end{cases}$$



四、观测量的理论

4.1 观测量作为标量

时空坐标 $\rightarrow$ 物理事件的时空位置和四个数构成的数组之间一一对应的关系

测量问题 ① 确定的观察对象 $\rightarrow$ 物理量

② 事件发生点的时空几何性质

③ 确定的参考系 $\rightarrow$ 按已知方式运动的观测者

④ 表述物理量和观测者运动的坐标系

$\rightarrow$ 观测量必定是一个标量。

4.2 时空间隔的测量和投影算符

观测者的运动由四维世界线刻画 $x_{\text{obs}}^{\mu} = x_{\text{obs}}^{\mu}(\tau)$

$\downarrow$ 切矢量为四维速度 $u^{\mu} = \frac{dx_{\text{obs}}^{\mu}}{d\tau}$ 归一为 $g_{\mu\nu} u^{\mu} u^{\nu} = -1$ 。
认为自己“静止” $\rightarrow$ 世界线的切向为时间轴方向
与时轴正交的三维子空间为普通空间

时空测量的基本观念：四维矢量 $\left\{ \begin{array}{l} \text{往观测者时间轴投影} \\ \text{往观测者空间部分投影} \end{array} \right.$

(1) 时间投影算符

$$\pi^{\mu\nu} \equiv -u^{\mu} u^{\nu}$$

A^{μ} 的投影 $A_{\parallel}^{\mu} = \pi_{\nu}^{\mu} A^{\nu}$

性质：① $\forall A^{\mu}$ ，投影后 $A_{\parallel}^{\mu}$ 平行于时轴 u^{μ}

$$A^{\mu} = \pi_{\nu}^{\mu} A^{\nu} = -u^{\mu} u_{\nu} A^{\nu} = -(u_{\nu} A^{\nu}) u^{\mu}$$

② $A^{\mu} \parallel u^{\mu}$ ，那么 $A_{\parallel}^{\mu} = A^{\mu}$

③ $\pi_{\nu}^{\mu} \pi_{\lambda}^{\nu} = \pi_{\lambda}^{\mu}$ 投影后再投影一次等于一次投影

$$\Delta\tau = \sqrt{-g_{\mu\nu} dx_{\text{obs}}^{\mu} dx_{\text{obs}}^{\nu}} = |g_{\mu\nu} u^{\mu} dx^{\nu}|$$

$\rightarrow$ $\left\{ \begin{array}{l} \text{时间间隔} \\ \text{空间间隔} \end{array} \right.$

(2) 空间投影算符

$$h^{\mu\nu} \equiv g^{\mu\nu} + u^{\mu} u^{\nu}$$

A^{μ} 的投影 $A_{\perp}^{\mu} = h_{\nu}^{\mu} A^{\nu}$

性质：① $A_{\perp}^{\mu} u_{\mu} = 0$

② $A^{\mu} \perp u^{\mu}$ ，那么 $A_{\perp}^{\mu} = A^{\mu}$

③ $A^{\mu} \parallel u^{\mu}$ ， $A_{\perp}^{\mu} = 0$

④ $A_{\perp}^{\mu} + A_{\parallel}^{\mu} = A^{\mu}$

⑤ $h_{\nu}^{\mu} h_{\lambda}^{\nu} = h_{\lambda}^{\mu}$

⑥ $h_{\nu}^{\mu} \pi_{\lambda}^{\nu} = \pi_{\lambda}^{\mu} h_{\lambda}^{\nu} = 0$



4.3 物理的坐标系

考虑特殊的观测者, 具有四维速度 $u^\alpha \neq 0$; $u^i = 0$

这种观测者相对坐标系局域静止/相对局域随动

$$u^\alpha \text{ 归一化 } u^\alpha = 1/\sqrt{-g_{00}}$$

适用的普通空间度规

$$h_{\alpha\beta} = g_{\alpha\beta} + u_\alpha u_\beta = g_{\alpha\beta} - \frac{(g_{00} g_{\beta 0})}{g_{00}}$$

$$\rightarrow h_{00} = h_{0i} = h_{i0} = 0, \quad h_{ij} = g_{ij} - (g_{i0} g_{j0})/g_{00}$$

$$\text{邻点距离公式 } \Delta L = \sqrt{h_{\alpha\beta} d\alpha^\alpha d\alpha^\beta} = \sqrt{h_{ij} d\alpha^i d\alpha^j}$$

→ 要求 $h_{ij} d\alpha^i d\alpha^j$ 为正定二次型

$$\hookrightarrow g_{00} < 0$$

$$|g_{00}| < 0$$

物理的坐标系条件

...

...

弯曲时空和平坦时空两点差别:

① 时间对准问题

② 有限距离问题

4.4 四轴系和局域测量的一般概念

②

参考系 $\xrightarrow{\text{世界线切向}}$ $\begin{cases} \text{世界线切向的一维} \rightarrow \text{时间} \\ \text{三维空间} \end{cases}$

↓ 引入三个两两正交的类空轴

四轴系——完整的局域参考系

观测者的世界线切向量 u^μ

$$\hookrightarrow \text{四轴系第零轴 } w_\alpha^\mu = u^\mu$$

$$\text{它是类时轴 归一为 } g_{\mu\nu} w_\alpha^\mu w_\beta^\nu = -1$$

$$\text{三个两两正交的类空轴 } w_a^\mu \quad (a=1,2,3)$$

$$\text{归一为 } g_{\mu\nu} w_a^\mu w_b^\nu = \begin{cases} 1, & a=b \\ 0, & a \neq b. \end{cases}$$

$$\text{同时有 } g_{\mu\nu} w_\alpha^\mu w_\beta^\nu = 0$$

$$\text{那么, 四轴系正交归一性为 } g_{\mu\nu} w_\alpha^\mu w_\beta^\nu = \eta_{\alpha\beta} \text{ 闵氏度规}$$

$$\text{第一重要性是完备性 } w_\alpha^\mu w_\beta^\nu = \delta_\beta^\alpha$$



4.5 两个观测者的测量的比较

两观测者 u, k , 有四维速 u^μ, k^μ

观测同一个事件间隔 dx^μ

$$u dx^\alpha = dx^\mu w_\mu^\alpha \rightarrow dx^\mu = u dx^\alpha w_\alpha^\mu$$

$$k dx^\alpha = dx^\mu A_\mu^\alpha \quad \leftarrow \alpha \rightarrow \beta$$

$$k dx^\alpha = u dx^\beta (w_\beta^\mu \lambda_\mu^\alpha)$$

$$A_\beta^\alpha = w_\beta^\mu \lambda_\mu^\alpha \quad \text{线性变换的变换矩阵}$$

仅取决于两个四轴系, 而不依赖于坐标系.

4.6 局域惯性系条件

$$\Omega_{ab} = 0 \quad (\dot{u}^\mu = 0)$$

沿测地线运动, 即在引力场中自由运动.

如果 $\Omega_{ab} = 0$

即三个空间轴都不转动, 这是费米平动参考系

结论是: 自由下落的费米平动参考系才构成一个局域的惯性系.



五. 球对称的引力场

5.1 球对称度规场的一般结构

空间球对称性 $\rightarrow$ 三个两两正交对称轴

(绕任一无穷小转动而不改变)

绕轴转动的映射 $\rightarrow$ 等度规映射
转动映射的生成元 $\rightarrow$ 凯林矢量场

取类笛卡尔坐标 (x^0, x^1, x^2, x^3)

度规为 $ds^2 = g_{00}(dx^0)^2 + 2g_{0i}dx^0dx^i + g_{ij}dx^i dx^j$

无穷小映射形式 $x^\mu = x^\mu + \varepsilon \xi^\mu$ ε 为转角

以绕 x^1 轴为例 $\xi^0 = \xi^1 = 0, \xi^2 = x^3, \xi^3 = -x^2$ (index)

即 $\xi_{\mu}^\mu = \delta_2^\mu x^3 - \delta_3^\mu x^2$

同理有 $\xi_{\mu}^\mu = \delta_3^\mu x^1 - \delta_1^\mu x^3, \xi_{\mu}^\mu = \delta_1^\mu x^2 - \delta_2^\mu x^1$

凯林方程有 $g_{\mu\nu,\rho}\xi^\rho + g_{\mu\rho}\xi_{,\nu}^\rho + g_{\nu\rho}\xi_{,\mu}^\rho = 0$

对于 g_{00} , 球对称性要求 $g_{00} = F(r, x^0), r = \sqrt{(x^1)^2 + (x^2)^2 + (x^3)^2}$

F 为任意函数

对于 g_{0i} , 要求 $g_{0i} = G(r, x^0)x^i$, G 是任意函数

对于 g_{ij} , $g_{ij} = A(r, x^0)\delta^{ij} + B(r, x^0)x^i x^j$

A, B 为任意函数

球对称度规的一般结构为

$$ds^2 = F(r, x^0)(dx^0)^2 + 2G(r, x^0)x^i dx^i dx^0 + A(r, x^0)dx^i dx^i + B(r, x^0)(x^i dx^i)^2$$

以球坐标 (r, θ, φ) 代替 (x^1, x^2, x^3) , 有

$$ds^2 = F(r, t)dt^2 + L(r, t)(d\theta^2 + \sin^2\theta d\varphi^2) + M(r, t)dr^2 + N(r, t)drdt$$

$\downarrow$ 两步坐标变换

$$ds^2 = -e^{\nu(r, t)}dt^2 + e^{\mu(r, t)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2) \rightarrow \text{充分化简后球对称场度规的一般形式}$$

$\mu(r, t)$ 与 $\nu(r, t)$ 由爱因斯坦场方程确定。

